

SHASHWAT RAJ

Applied AI Engineer

@ shashwatrajiitd@gmail.com

📍 Mumbai, Maharashtra

🌐 shashwatrajiitd

🌐 portfolio-web-qtsas2ligq-el.a.run.app



WORK EXPERIENCE

Applied AI Engineer

Purple.com (Manash Lifestyle Pvt. Ltd.)

📅 May 2025 – Present

📍 Mumbai, India

GenAI Creative Automation Platform

- **Built & deployed an autonomous multi-agent system** for large-scale creative generation across paid marketing channels (Meta, Google Ads).
- **Phase 1 – Delivered system design & architecture:**
 - **Ideation Agents:** Generate high-volume campaign directions and ad concepts.
 - **Generation Agents:** Produce high-fidelity static creatives via orchestrated LLM/VLM and diffusion pipelines.
 - **Moderation Agents:** Rank and filter outputs for brand compliance using vision-based evaluators.
 - **Outcome:** Generating 3000+ marketing ideas and 1500+ creatives a month across 6 Brands, 30+ SKUs and 25+ campaigns.
- **Impact:** 10–15× faster creative production (1-2 weeks → < 3 hours); ~95% cost reduction (INR 1500–2000 → < INR 50 per asset); enabled 10-15× more creative variants per campaign.
- **Phase 2 – In execution:**
 - **Platform & surface expansion:** Scaling the platform to power L1 app surfaces including widgets, banners, category & theme-based recommendations.
 - **Centralized creative infrastructure:** Building a Creative Generation Service & an org-wide Master Content Bank (MCB) to eliminate creative bottlenecks & enable structured, reusable assets at scale.
- **Phase 3 – Roadmap & Ownership:**
 - **AI-powered personalization and content orchestration:** User-level widget personalization, smart shuffling using personas and behavioral signals, and GenAI-driven PDP image and video creative automation.
 - **Scalable governance & control:** Enforcement of business logic, compliance requirements, & brand guidelines across all automated experiences.
- **Tech stack:** Gemini, OpenAI, Stable Diffusion (LoRA fine-tuning), LLMs/VLMs, vision systems, microservices, orchestration, and MLOps.

INTERNSHIP EXPERIENCE

Software Developer Intern

Samsung R&D

📅 May 2024 – July 2024

📍 Bangalore, India

Real-Time Revenue Forecasting and Anomaly Detection Platform

- Built a **real-time revenue forecasting and anomaly detection pipeline** for Samsung Ads using regression and transformer-based time-series models.
- Implemented **residual-based anomaly detection** to flag near-real-time revenue deviations and improve system observability.
- **Impact:** Increased forecast accuracy, reduced detection latency, and enabled reliable inference at high data throughput.
- **Tech stack:** Python, ML models, Apache Kafka, AWS, distributed data processing.

ABOUT

Applied AI Engineer specializing in GenAI systems, with a strong foundation in Mathematics and Computing from IIT Delhi, focused on building scalable, business-driven AI solutions.

EDUCATION

B.Tech. in Mathematics & Computing

Indian Institute of Technology, Delhi

📅 2021 – 2025

📍 New Delhi, India

Specialisation in Machine Learning & Applied AI

High School (12th Grade)

Stephens International Public School

📅 2019 – 2021

📍 Jammu, India

ACHIEVEMENTS



JEE Advanced & Mains 2021

Secured 99.89 percentile nationwide among 1M + candidates



NTSE Scholar 2019

Awarded National Talent Scholarship among 1M + students



Samsung SWC Test

Qualified Samsung's Advanced Software Competency Test on first attempt



Regional Mathematical Olympiad

Received certificate of merit twice for mathematical excellence

TECHNICAL SKILLS

Programming & Systems

Python

SQL

C/C++

AWS

JavaScript

TypeScript

DSA

GCP

Git

API Design

Object-Oriented Design

Docker

Kafka

GenAI, ML & AI Models

LLM/VLM

Diffusion Models

OpenAI

RAG

Agentic Systems

Prompt Orchestration

LoRA Fine-Tuning

Vector DB

Google Gemini